Geometric Intuition and Formalism for Lagrange Multipliers

Introduction

In this section we shall look at some constrained optimization problems. The method of Lagrange multipliers allows us to transform constrained problems into unconstrained ones through the use of auxiliary variables. We shall focus on the two dimensional case.

Gradients and Directional Derivatives

For a differentiable scalar function $f(x, y)$, the **gradient** is the vector of partial derivatives:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

If $\mathbf{u}$ is a unit vector, the **directional derivative** of f in the direction $\mathbf{u}$ is

$$D_{\mathbf{u}}[f] = \nabla f \cdot \mathbf{u}.$$

Level Curves

Every point (x, y) in the domain of f lies on a **level curve** defined by $f(x, y) = k$ for some constant $k = f(x, y)$. If $\mathbf{v}$ is a vector tangent to the level curve at that point, then the directional drivative in the direction of $\mathbf{v}$ vanishes.

$$\nabla f \cdot \mathbf{v} = 0,$$

This implies that ∇f is perpendicular to the level curve $f(x, y) = k$. In other words, the gradient points in the direction of steepest ascent and is normal to the contours of constant f.

Stop and think about this as it is key for everything that follows.

Tangency Condition for Constrained Extrema

Suppose we wish to maximize $f(x, y)$ subject to constraint $g(x, y) = c$. At an extremum (x^*, y^*) the level curve $f(x, y) = f(x^*, y^*)$ must be tangent to the constraint curve $g(x, y) = c$. This implies that both gradients ∇f and ∇g are perpendicular to their shared tangent line at (x^*, y^*). Consequently, the gradients must be parallel:

$$\nabla f = \lambda \nabla g \quad \text{for some scalar } \lambda.$$

If the gradients were not parallel, movement along the constraint would increase or decrease f, contradicting optimality.

Illustrative Example

Maximize $f(x, y) = x + y$ subject to $g(x, y) = x^2 + y^2 = 1$: From our previous discussion the solution must occur where the line is tangent to the unit circle.

Consider the point $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$ which lies on the unit circle. We shall show that this is the maximum. Firstly we have::

$$\nabla f = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla g = \begin{bmatrix} 2x \\ 2y \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

That is $\nabla f = \lambda \nabla g$ holds with $\lambda = \frac{1}{\sqrt{2}}$, confirming tangency. And that's all we have to prove!

Tangency of level curves $\implies \nabla f \parallel \nabla g \implies$ Lagrange multiplier condition

The Lagrangian Framework

The geometric condition $\nabla f = \lambda \nabla g$ is necessary but insufficient—it doesn't enforce $g(x, y) = c$. We resolve this by constructing the **Lagrangian**:

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda \left(g(x, y) - c \right).$$

This single function encodes both requirements for extrema.

Why the Lagrangian Works

The critical points of $\mathcal{L}$ satisfy two conditions:

1. **Gradient alignment:** $\dfrac{\partial \mathcal{L}}{\partial x} = 0$ and $\dfrac{\partial \mathcal{L}}{\partial y} = 0$ yield
$$\nabla f - \lambda \nabla g = \mathbf{0} \quad \implies \quad \nabla f = \lambda \nabla g.$$
2. **Constraint satisfaction:** $\dfrac{\partial \mathcal{L}}{\partial \lambda} = 0$ gives
$$-(g(x, y) - c) = 0 \quad \implies \quad g(x, y) = c.$$

Solving $\nabla \mathcal{L} = \mathbf{0}$ simultaneously enforces both conditions, identifying candidate extrema.

The Lagrangian $\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - c)$ unifies:
1. Gradient alignment via $\partial \mathcal{L}/\partial x = \partial \mathcal{L}/\partial y = 0$
2. Constraint satisfaction via $\partial \mathcal{L}/\partial \lambda = 0$

Worked Examples

Example 1: Maximizing Cylinder Volume

Problem: Find radius r and height h maximizing volume $V = \pi r^2 h$ for a closed cylinder with fixed surface area $S = 12\pi$.

Solution:

- **Constraint:** $2\pi r^2 + 2\pi r h = 12\pi$ (top, bottom, and lateral surface)
- **Lagrangian:** $\mathcal{L}(r, h, \lambda) = \pi r^2 h - \lambda(2\pi r^2 + 2\pi r h - 12\pi)$
- **First-order conditions:**
$$\frac{\partial \mathcal{L}}{\partial r} = 2\pi r h - \lambda(4\pi r + 2\pi h) = 0 \qquad (1)$$
$$\frac{\partial \mathcal{L}}{\partial h} = \pi r^2 - \lambda(2\pi r) = 0 \qquad (2)$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2\pi r^2 + 2\pi r h - 12\pi) = 0 \qquad (3)$$
- **Solving:**
  - From (2): $\pi r^2 = 2\pi r \lambda \implies \lambda = r/2$ (assuming $r \neq 0$)
  - Substitute into (1): $2\pi r h = (r/2)(4\pi r + 2\pi h) \implies 4h = 4r + 2h \implies h = 2r$
  - Substitute into (3): $2\pi r^2 + 2\pi r(2r) = 12\pi \implies 6\pi r^2 = 12\pi \implies r = \sqrt{2}$
  - Thus $h = 2\sqrt{2}$, and maximum volume $V = \pi(\sqrt{2})^2(2\sqrt{2}) = 4\sqrt{2}\pi$
  - **Interpretation:** The optimal cylinder has height equal to diameter ($h = 2r$).

Example 2: Minimum Distance to a Plane

Problem: Find the point on plane $2x + 3y + z = 6$ closest to the origin.

Solution:

- **Objective:** Minimize squared distance $f(x, y, z) = x^2 + y^2 + z^2$ (avoids square roots)
- **Lagrangian:** $\mathcal{L}(x, y, z, \lambda) = x^2 + y^2 + z^2 - \lambda(2x + 3y + z - 6)$
- **First-order conditions:**

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - 2\lambda = 0 \implies x = \lambda \qquad (1)$$
$$\frac{\partial \mathcal{L}}{\partial y} = 2y - 3\lambda = 0 \implies y = \tfrac{3}{2}\lambda \qquad (2)$$
$$\frac{\partial \mathcal{L}}{\partial z} = 2z - \lambda = 0 \implies z = \tfrac{\lambda}{2} \qquad (3)$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2x + 3y + z - 6) = 0 \qquad (4)$$

- **Solving:**
  - Substitute (1)-(3) into (4): $2(\lambda) + 3(\frac{3}{2}\lambda) + \frac{\lambda}{2} = 6 \implies 7\lambda = 6 \implies \lambda = \frac{6}{7}$
  - Critical point: $\left(\frac{6}{7}, \frac{9}{7}, \frac{3}{7}\right)$
  - Minimum distance: $\sqrt{\left(\frac{6}{7}\right)^2 + \left(\frac{9}{7}\right)^2 + \left(\frac{3}{7}\right)^2} = \sqrt{\frac{126}{49}} = \frac{3\sqrt{14}}{7}$
  - **Verification:** The vector $\langle \frac{6}{7}, \frac{9}{7}, \frac{3}{7} \rangle$ is parallel to the plane's normal $\langle 2, 3, 1 \rangle$, confirming geometric intuition.

Example 3: Cobb-Douglas Production Maximization

Problem: Maximize output $Q = K^{1/3}L^{2/3}$ subject to budget constraint $2L + K = 12$, where K=capital, L=labor.

Solution:

- **Lagrangian:** $\mathcal{L}(K, L, \lambda) = K^{1/3}L^{2/3} - \lambda(2L + K - 12)$
- **First-order conditions:**

$$\frac{\partial \mathcal{L}}{\partial K} = \tfrac{1}{3}K^{-2/3}L^{2/3} - \lambda = 0 \qquad (1)$$
$$\frac{\partial \mathcal{L}}{\partial L} = \tfrac{2}{3}K^{1/3}L^{-1/3} - 2\lambda = 0 \qquad (2)$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2L + K - 12) = 0 \qquad (3)$$

- **Solving:**
- Divide (1) by (2): $\dfrac{\frac{1}{3}K^{-2/3}L^{2/3}}{\frac{2}{3}K^{1/3}L^{-1/3}} = \dfrac{\lambda}{2\lambda} \implies \frac{1}{2} \cdot \frac{L}{K} = \frac{1}{2} \implies L = K$
- Substitute into (3): $2K + K = 12 \implies K = 4$, $L = 4$
- Maximum output: $Q = 4^{1/3} \cdot 4^{2/3} = 4$
- **Economic insight:** Optimal allocation equalizes capital and labor investments despite different input prices.

Conclusion

Lagrange multipliers transform constrained optimization into a system of equations by leveraging geometric principles: extrema occur where objective and constraint gradients are parallel. The Lagrangian formalism unifies this gradient condition with constraint satisfaction. This approach generalizes to:

- Multiple constraints (introduce λ_i for each $g_i(\mathbf{x}) = c_i$)
- Higher dimensions (gradients remain normal to level surfaces)
- Inequality constraints (via Karush-Kuhn-Tucker conditions) The geometric insight—that solutions occur at tangency points—provides an intuitive anchor for these powerful techniques.